

# Research Notes on Ensemble learning

## Ensemble learning

### >> Section 1

Bayesian model averaging Bayesian model averaging (BMA) makes predictions by averaging the predictions of models weighted by their posterior probabilities given the data. BMA is known to generally give better answers than a single model, obtained, e.g., via stepwise regression, especially where very different models have nearly identical performance in the training set but may otherwise perform quite differently. The question with any use of Bayes' theorem is the prior, i.e., the probability (perhaps subjective) that each model is the best to use for a given purpose. Conceptually, BMA can be used with any prior. R packages ensembleBMA and BMA use the prior implied by the Bayesian information criterion, (BIC), following Raftery (1995). R package BAS supports the use of the priors implied by Akaike information criterion (AIC) and other criteria over the alternative models as well as priors over the coefficients. The difference between BIC and AIC is the strength of preference for parsimony. BIC's penalty for model complexity is  $\ln(n)k$ , while AIC's is  $2k$ . Large-sample asymptotic theory establishes that if there is a best model, then with increasing sample sizes, BIC is strongly consistent, i.e., will almost certainly find it, while AIC may not, because AIC may continue to place excessive posterior probability on models that are more complicated than they need to be. On the other hand, AIC and AICc are asymptotically "efficient" (i.e., minimum mean square prediction error), while BIC is not. Haussler et al. (1994) showed that when BMA is used for classification, its expected error is at most twice the expected error of the Bayes optimal classifier. Burnham and Anderson (1998, 2002) contributed greatly to introducing a wider audience to the basic ideas of Bayesian model averaging and popularizing the methodology. The availability of software, including other free open-source packages for R beyond those mentioned above, helped make the methods accessible to a wider audience.

### >> Section 2

Malware Detection Classification of malware codes such as computer viruses, computer worms, trojans, ransomware and spywares with the usage of machine learning techniques, is inspired by the document categorization problem. Ensemble learning systems have shown a proper efficacy in this area. Model Hardening Malware detection models, like all machine learning models, are vulnerable to adversarial machine learning attacks where an attacker pushes the boundary of what is classified as malware. By rotating through an ensemble of models using moving target defense, the attacker has a diminished knowledge advantage.

### >> Section 3

Boosting involves training successive models by emphasizing training data mis-classified by previously learned models. Initially, all data (D1) has equal weight and is used to learn a base model M1. The examples mis-classified by M1 are assigned a weight greater than correctly classified examples. This boosted data (D2) is used to train a second base model M2, and so on. Inference is done by voting. In some cases, boosting has yielded better accuracy than bagging, but tends to over-fit more. The most common implementation of boosting is Adaboost, but some newer algorithms are reported to achieve better results.

### >> Section 4

Bucket of models A "bucket of models" is an ensemble technique in which a model selection algorithm is used to choose the best model for each problem. When tested with only one problem, a bucket of models can produce no better results than the best model in the set, but when evaluated across many problems, it will typically produce much better results, on average, than any model in the set. The most common approach used for model-selection is cross-validation selection (sometimes called a "bake-off contest"). It is described with the following pseudo-code:

### >> Section 5

For each model m in the bucket: Do c times: (where 'c' is some constant) Randomly divide the training dataset into two sets: A and B Train m with A Test m with B Select the model that obtains the highest average score

### >> Section 6

In statistics and machine learning, ensemble methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone. Unlike a statistical ensemble in statistical mechanics, which is usually infinite, a machine learning ensemble consists of only a concrete finite set of alternative models, but typically allows for much more flexible structure to exist among those alternatives.

### >> Section 7

Overview Supervised learning algorithms search through a hypothesis space to find a suitable hypothesis that will make good predictions with a particular problem. Even if this space contains hypotheses that are very well-suited for a particular problem, it may be very difficult to find a good one. Ensembles combine multiple hypotheses to form one which should be theoretically better. Ensemble learning trains two or more machine learning algorithms on a specific classification or regression task. The algorithms

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within the ensemble model are generally referred as "base models", "base learners", or "weak learners" in literature. These base models can be constructed using a single modelling algorithm, or several different algorithms. The idea is to train a diverse set of weak models on the same modelling task, such that the outputs of each weak learner have poor predictive ability (i.e., high bias), and among all weak learners, the outcome and error values exhibit high variance. Fundamentally, an ensemble learning model trains at least two high-bias (weak) and high-variance (diverse) models to be combined into a better-performing model. The set of weak models — which would not produce satisfactory predictive results individually — are combined or averaged to produce a single, high performing, accurate, and low-variance model to fit the task as required. Ensemble learning typically refers to bagging (bootstrap aggregating), boosting or stacking/blending techniques to induce high variance among the base models. Bagging creates diversity by generating random samples from the training observations and fitting the same model to each different sample — also known as homogeneous parallel ensembles. Boosting follows an iterative process by sequentially training each base model on the up-weighted errors of the previous base model, producing an additive model to reduce the final model errors — also known as sequential ensemble learning. Stacking or blending consists of different base models, each trained independently (i.e. diverse/high variance) to be combined into the ensemble model — producing a heterogeneous parallel ensemble. Common applications of ensemble learning include random forests (an extension of bagging), Boosted Tree models, and Gradient Boosted Tree Models. Models in applications of stacking are generally more task-specific — such as combining clustering techniques with other parametric and/or non-parametric techniques. Evaluating the prediction of an ensemble typically requires more computation than evaluating the prediction of a single model. In one sense, ensemble learning may be thought of as a way to compensate for poor learning algorithms by performing a lot of extra computation. On the other hand, the alternative is to do a lot more learning with one non-ensemble model. An ensemble may be more efficient at improving overall accuracy for the same increase in compute, storage, or communication resources by using that increase on two or more methods, than would have been improved by increasing resource use for a single method. Fast algorithms such as decision trees are commonly used in ensemble methods (e.g., random forests), although slower algorithms can benefit from ensemble techniques as well. By analogy, ensemble techniques have been used also in unsupervised learning scenarios, for example in consensus clustering or in

anomaly detection.